MKRYPT CYBERSECURITY DETECTION SERIES – BEGINNER

# Top 20 Production KQL Detection Rules Runbook

A standardized engineering handbook containing production-grade Kusto Query Language (KQL) detection logic, visual attack architecture diagrams, MITRE ATT&CK mappings, and analyst triage workflows for onboarding new enterprise SOC environments.

| | |
|---|---|
| **Target Platform:** | Microsoft Sentinel / Microsoft Defender XDR / Azure Data Explorer |
| **Framework Alignment:** | MITRE ATT&CK Enterprise Matrix (Tactics TA0001 – TA0040) |
| **Target Audience:** | SOC Analysts (L1/L2/L3), Detection Engineers, Threat Hunters |
| **Series Edition:** | Foundational to Advanced Enterprise Detections |

# Executive Overview & Master Detection Matrix

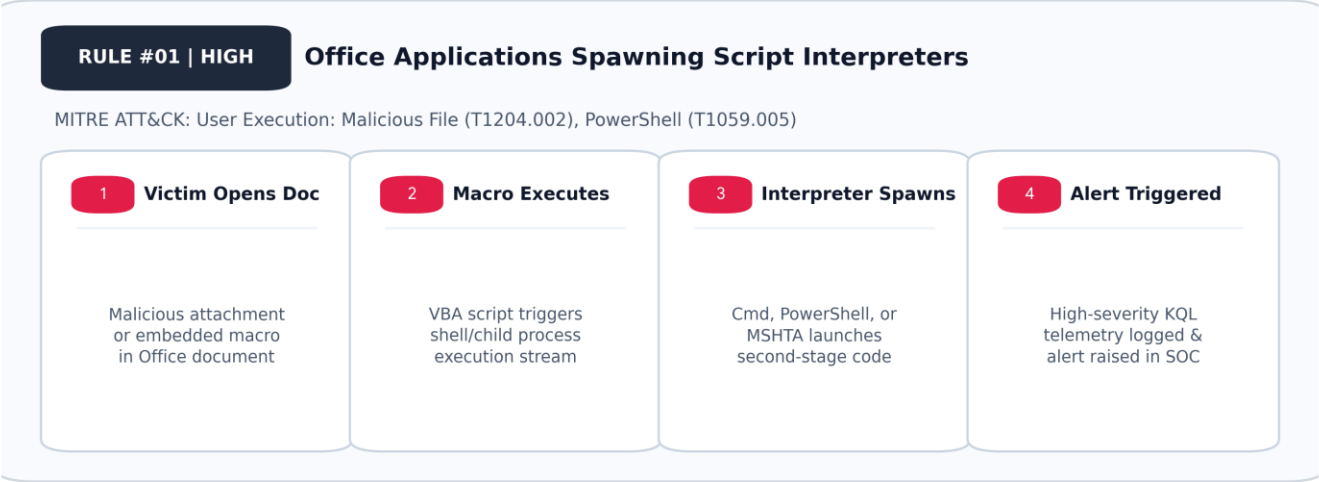
This runbook provides 20 foundational detection rules structured with dedicated single-page layouts. Each page includes formal detection metadata, visual kill-chain attack diagrams, copy-ready KQL queries, false-positive baselining advice, and standard operating procedures (SOP) for triage.

| # | Detection Title | Primary Tactic & Technique | Severity |
|---|---|---|---|
| 01 | Office Applications Spawning Script Interpreters | Execution | High |
| 02 | Certutil Payload Download & Decoding (LOLBin) | Defense Evasion, Ingress Tool Transfer | High |
| 03 | PowerShell Base64 Encoded Execution | Execution, Defense Evasion | Medium |
| 04 | Windows Script Host External Network Connection | Execution, Command and Control | High |
| 05 | Scheduled Task Creation via Native Command Line | Persistence, Execution | Medium |
| 06 | Persistence via Registry Run / RunOnce Keys | Persistence | High |
| 07 | New Service Installation Executing from User/Temp Paths | Persistence, Privilege Escalation | High |
| 08 | Privileged Local/Domain Group Membership Manipulation | Privilege Escalation, Persistence | High |
| 09 | Windows Security/System Event Log Clearance | Defense Evasion | High |
| 10 | Tampering with Antivirus & Defender Controls | Defense Evasion | High |
| 11 | Masquerading: Core System Binaries in Non-Standard Paths | Defense Evasion | High |
| 12 | Volume Shadow Copy Deletion (Ransomware Precursor) | Impact | Critical |
| 13 | LSASS OpenProcess Handle Scrape for Credential Dumping | Credential Access | High |
| 14 | Password Spraying Anomaly Across Multiple User Accounts | Credential Access | Medium |
| 15 | Rapid Sequence of Active Directory Reconnaissance Utilities | Discovery | Medium |
| 16 | Lateral Movement via PsExec Execution Artifacts | Lateral Movement, Execution | High |
| 17 | Remote Task Scheduling via Native Schtasks Commands | Lateral Movement, Persistence | High |
| 18 | Entra ID Global Administrator / High Privileged Role Assignment | Privilege Escalation, Persistence | Critical |
| 19 | OAuth Application Secret or Certificate Injection | Persistence, Defense Evasion | High |
| 20 | Outbound Beaconing to Dynamic DNS / Tunneling Infrastructure | Command and Control | Medium |

# RULE #01 | Office Applications Spawning Script Interpreters

| | | | |
|---|---|---|---|
| MITRE TACTIC | Execution | MITRE TECHNIQUE | T1059.005 |
| LOG SOURCE / TABLE | DeviceProcessEvents | ALERT SEVERITY | High |

**Threat Scenario & Context:** Detects Microsoft Office executables (Word, Excel, Outlook) spawning command shells or scripting interpreters such as cmd, powershell, wscript, or certutil, which is a classic macro/exploit weaponization vector.




### Production KQL Detection Logic

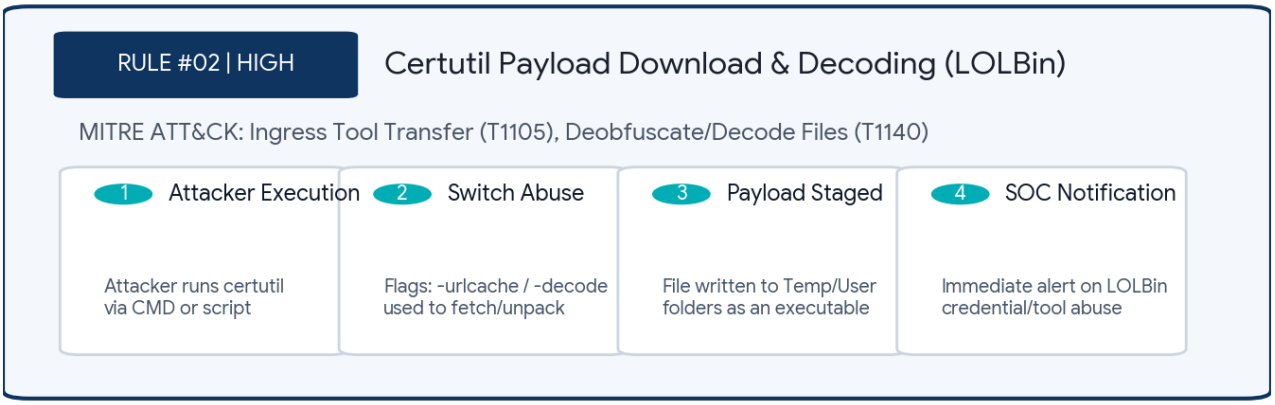
```
// Alert: Office App Spawning Script Interpreter
DeviceProcessEvents
| where InitiatingProcessFileName in~ (
    "winword.exe", "excel.exe", "powerpnt.exe", "outlook.exe"
  )
| where FileName in~ (
    "cmd.exe", "powershell.exe", "pwsh.exe", "wscript.exe",
    "cscript.exe", "mshta.exe", "certutil.exe", "bitsadmin.exe"
  )
| project TimeGenerated, DeviceName, AccountName, FileName,
          ProcessCommandLine, InitiatingProcessFileName,
          InitiatingProcessCommandLine
| extend MitreTactic = "Execution", MitreTechnique = "T1204.002"
```

| | |
|---|---|
| **False Positives & Tuning** | Legitimate enterprise add-ins or macros running custom scripts. Add exclusions by parent command line and digital signing status. |
| **SOC Triage & Response** | 1. Isolate endpoint immediately.<br>2. Inspect InitiatingProcessCommandLine and dropped email attachment.<br>3. Collect network connections made by child process. |

# RULE #02 | Certutil Payload Download & Decoding (LOLBin)

| | | | |
|---|---|---|---|
| MITRE TACTIC | Defense Evasion, Ingress Tool Transfer | MITRE TECHNIQUE | T1140 |
| LOG SOURCE / TABLE | DeviceProcessEvents | ALERT SEVERITY | High |

**Threat Scenario & Context:** Detects certutil.exe invoked with switches such as -urlcache or -decode, commonly abused to download second-stage malware or unpack obfuscated scripts under a trusted Microsoft binary.




### Production KQL Detection Logic

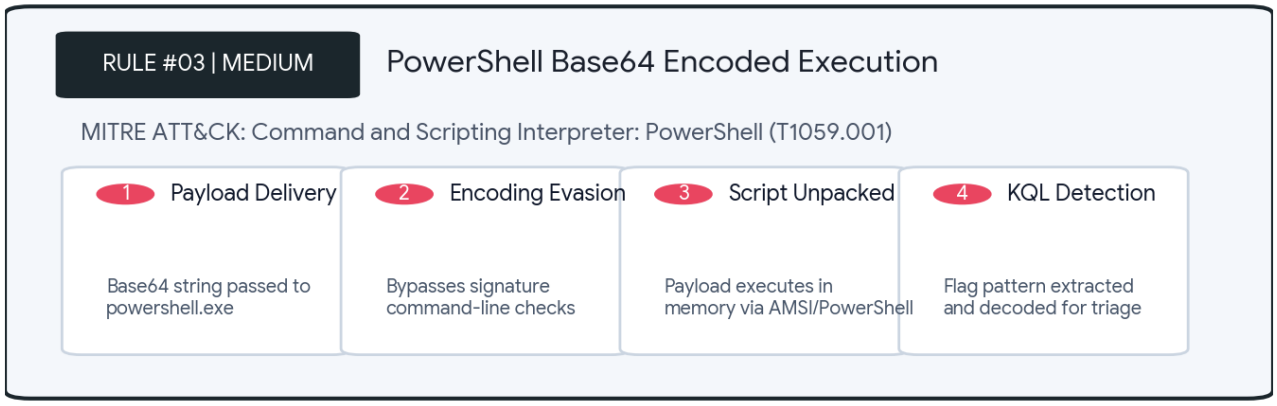
```
// Alert: Certutil Download or Base64 Decode
DeviceProcessEvents
| where FileName =~ "certutil.exe"
    or ProcessVersionInfoOriginalFileName =~ "certutil.exe"
| where ProcessCommandLine has_any (
    "-urlcache", "/urlcache", "-split", "/split", "-decode", "/decode"
  )
| project TimeGenerated, DeviceName, AccountName, ProcessCommandLine,
          InitiatingProcessCommandLine
| extend MitreTactic = "DefenseEvasion", MitreTechnique = "T1140"
```

| | |
|---|---|
| **False Positives & Tuning** | Automated enterprise certificate provisioning scripts decoding base64 certificates. Filter on specific system service accounts. |
| **SOC Triage & Response** | 1. Identify the URL or file path in ProcessCommandLine.<br>2. Submit target payload to sandbox.<br>3. Review initiating parent process tree. |

# RULE #03 | PowerShell Base64 Encoded Execution

| | | | |
|---|---|---|---|
| MITRE TACTIC | Execution, Defense Evasion | MITRE TECHNIQUE | T1059.001 |
| LOG SOURCE / TABLE | DeviceProcessEvents | ALERT SEVERITY | Medium |

**Threat Scenario & Context:** Identifies PowerShell instances spawned with encoded command arguments (-enc, -encodedcommand), commonly employed by cyber threat actors to bypass static command-line inspection and string matching.




### Production KQL Detection Logic

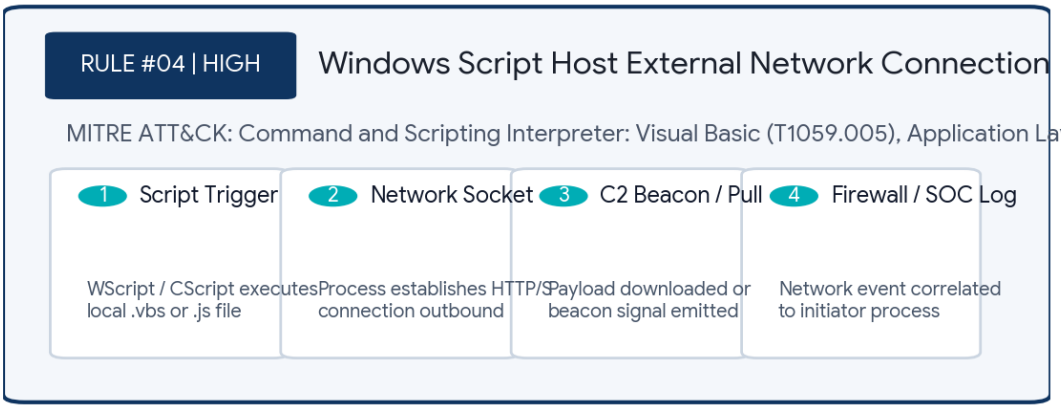
```
// Alert: PowerShell Encoded Execution
DeviceProcessEvents
| where FileName in~ ("powershell.exe", "pwsh.exe")
| where ProcessCommandLine has_any (
    "-e ", "-enc ", "-encodedcommand ", "/e ", "/enc "
  )
| project TimeGenerated, DeviceName, AccountName, ProcessCommandLine,
          InitiatingProcessFileName
| extend MitreTactic = "Execution", MitreTechnique = "T1059.001"
```

| | |
|---|---|
| **False Positives & Tuning** | Legitimate enterprise administration scripts (Microsoft Exchange Management, SCCM, Intune scripts). |
| **SOC Triage & Response** | 1. Decode the Base64 command using CyberChef.<br>2. Verify initiating parent process.<br>3. Inspect child network connections and script block logs (EID 4104). |

# RULE #04 | Windows Script Host External Network Connection

| | | | |
|---|---|---|---|
| MITRE TACTIC | Execution, Command and Control | MITRE TECHNIQUE | T1071 |
| LOG SOURCE / TABLE | DeviceNetworkEvents | ALERT SEVERITY | High |

**Threat Scenario & Context:** Monitors wscript.exe and cscript.exe making successful outbound network connections to public IP addresses, a primary indicator of VBScript/JScript staging droppers or beaconing activity.




### Production KQL Detection Logic

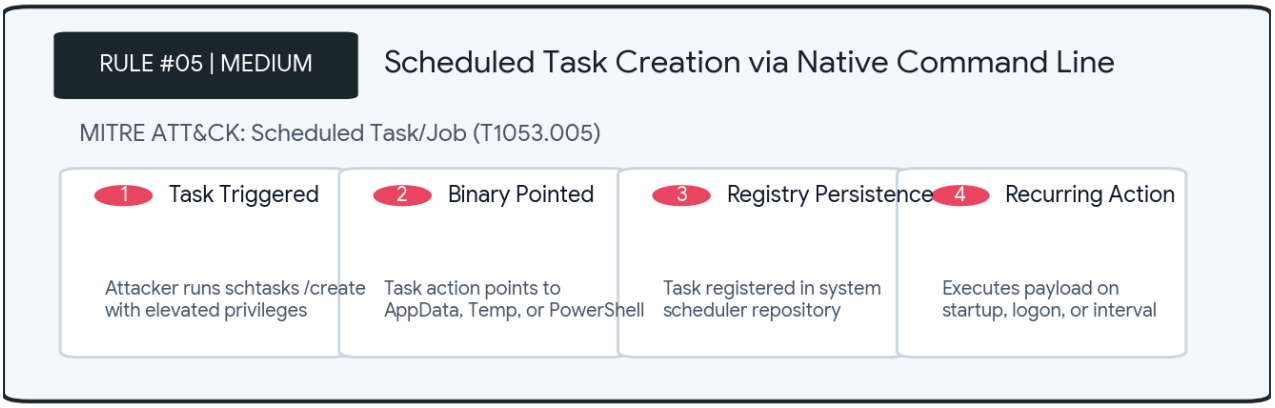
```
// Alert: Wscript/Cscript Outbound Connection
DeviceNetworkEvents
| where InitiatingProcessFileName in~ ("wscript.exe", "cscript.exe")
| where ActionType == "ConnectionSuccess"
| where RemoteIPType != "Private" and RemotePort in (80, 443, 8080, 8443)
| project TimeGenerated, DeviceName, InitiatingProcessFileName,
        RemoteIP, RemoteUrl, RemotePort
| extend MitreTactic = "Execution", MitreTechnique = "T1059.005"
```

| | |
|---|---|
| **False Positives & Tuning** | Very rare in well-configured environments. Legacy enterprise IT logon scripts may communicate with external telemetry. |
| **SOC Triage & Response** | 1. Check reputation of RemoteIP and RemoteUrl on VirusTotal/Threat Intel.<br>2. Isolate host.<br>3. Retrieve script file referenced in process history. |

# RULE #05 | Scheduled Task Creation via Native Command Line

| | | | |
|---|---|---|---|
| MITRE TACTIC | Persistence, Execution | MITRE TECHNIQUE | T1053.005 |
| LOG SOURCE / TABLE | DeviceProcessEvents | ALERT SEVERITY | Medium |

**Threat Scenario & Context:** Detects command-line creation or modification of Windows Scheduled Tasks using schtasks.exe pointing to scripting shells, user directories, or suspicious staging paths.




### Production KQL Detection Logic

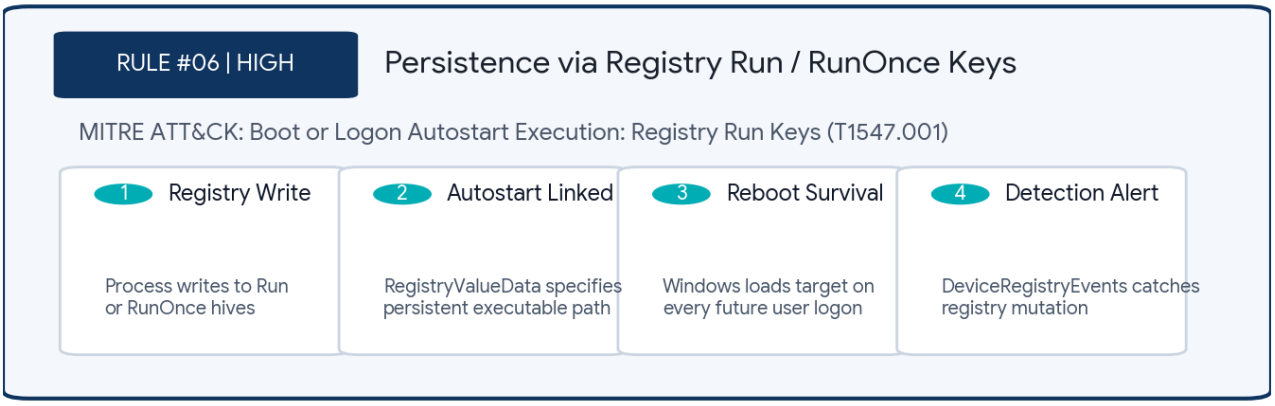
```
// Alert: Schtasks Creation or Modification
DeviceProcessEvents
| where FileName =~ "schtasks.exe"
| where ProcessCommandLine has_any ("/create", "/change")
| where ProcessCommandLine has_any (
    "powershell", "cmd", "wscript", "rundll32",
    "regsvr32", "AppData", "Temp"
  )
| project TimeGenerated, DeviceName, AccountName, ProcessCommandLine,
          InitiatingProcessCommandLine
| extend MitreTactic = "Persistence", MitreTechnique = "T1053.005"
```

| | |
|---|---|
| **False Positives & Tuning** | Application updaters (Google Chrome, Edge, Zoom, Adobe) creating auto-update tasks. Baseline by vendor digital signatures. |
| **SOC Triage & Response** | 1. Parse task name (/tn) and program (/tr).<br>2. Inspect author and run permissions.<br>3. Delete suspicious task using schtasks /delete if malicious. |

# RULE #06 | Persistence via Registry Run / RunOnce Keys

| | | | |
|---|---|---|---|
| MITRE TACTIC | Persistence | MITRE TECHNIQUE | T1547.001 |
| LOG SOURCE / TABLE | DeviceRegistryEvents | ALERT SEVERITY | High |

**Threat Scenario & Context:** Tracks writes or modifications to Registry Run or RunOnce keys pointing to executable binaries, scripts, or LOLBins designed to maintain persistence across user logons and reboots.




### Production KQL Detection Logic

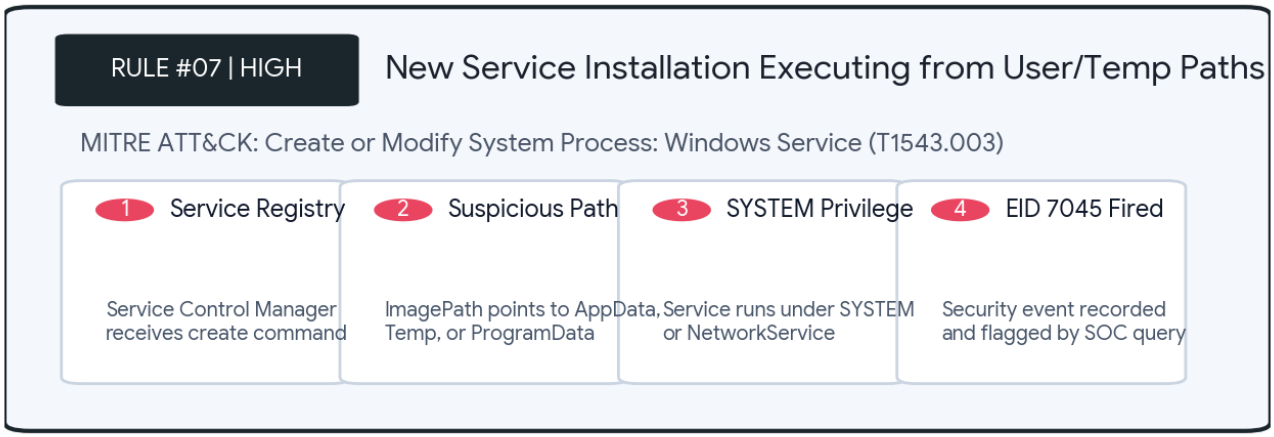
```
// Alert: Run Key Value Added/Modified
DeviceRegistryEvents
| where ActionType in ("RegistryValueSet", "RegistryKeyCreated")
| where RegistryKey has_any (@"\CurrentVersion\Run", @"\CurrentVersion\RunOnce")
| where RegistryValueData has_any (
    ".exe", ".dll", ".ps1", ".vbs", ".bat", "powershell", "wscript"
  )
| project TimeGenerated, DeviceName, RegistryKey, RegistryValueName,
          RegistryValueData, InitiatingProcessFileName
| extend MitreTactic = "Persistence", MitreTechnique = "T1547.001"
```

| | |
|---|---|
| **False Positives & Tuning** | Legitimate newly installed software registering system tray utilities or startup helpers. |
| **SOC Triage & Response** | 1. Verify vendor and signature of the file in RegistryValueData.<br>2. Review hash in threat intelligence feeds.<br>3. Remove rogue registry key. |

# RULE #07 | New Service Installation Executing from User/Temp Paths

| | | | |
|---|---|---|---|
| MITRE TACTIC | Persistence, Privilege Escalation | MITRE TECHNIQUE | T1543.003 |
| LOG SOURCE / TABLE | SecurityEvent | ALERT SEVERITY | High |

**Threat Scenario & Context:** Detects the creation of Windows services (Event ID 7045) where the binary path references user-writable or temporary directories (e.g., AppData, Temp, PerfLogs).




### Production KQL Detection Logic

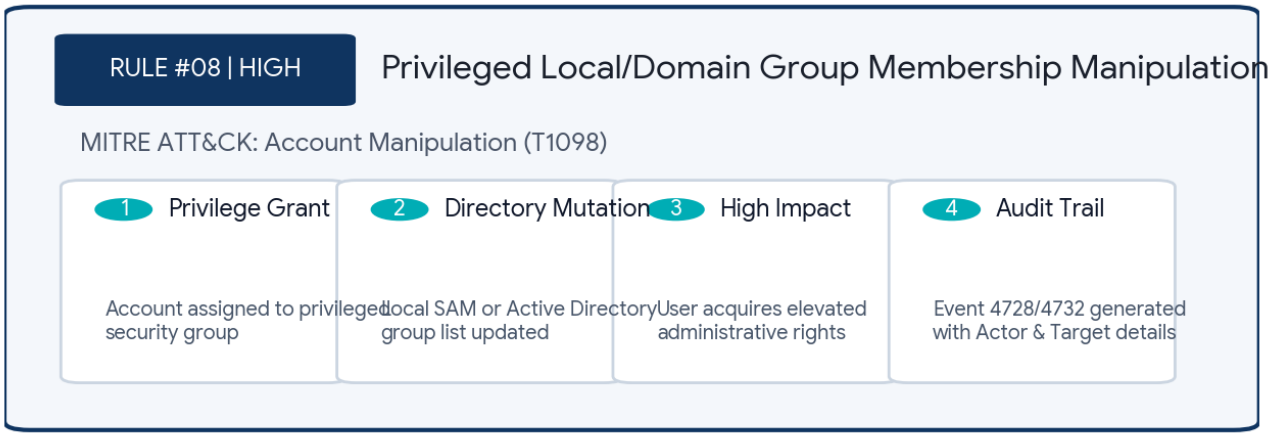
```
// Alert: Service Installed Executing from User/Temp Paths
SecurityEvent
| where EventID == 7045
| extend ServicePath = tostring(EventData.ImagePath)
| where ServicePath has_any (
    @"C:\Users\", @"\AppData\", @"\Temp\",
    @"\PerfLogs\", @"C:\ProgramData\"
  )
| project TimeGenerated, Computer, ServiceName = EventData.ServiceName,
          ServicePath, Account
| extend MitreTactic = "Persistence", MitreTechnique = "T1543.003"
```

| | |
|---|---|
| **False Positives & Tuning** | Developer utilities, portable tools, or uninstaller stubs executing from temporary locations during software deployment. |
| **SOC Triage & Response** | 1. Check ServiceName and Account that initiated creation.<br>2. Scan ServicePath executable.<br>3. Stop and delete service with sc.exe stop / sc.exe delete. |

# RULE #08 | Privileged Local/Domain Group Membership Manipulation

| | | | |
|---|---|---|---|
| MITRE TACTIC | Privilege Escalation, Persistence | MITRE TECHNIQUE | T1098 |
| LOG SOURCE / TABLE | SecurityEvent | ALERT SEVERITY | High |

**Threat Scenario & Context:** Detects addition of users to sensitive local or domain administrative groups (such as Administrators, Domain Admins, Remote Desktop Users) using Event IDs 4728 and 4732.




### Production KQL Detection Logic

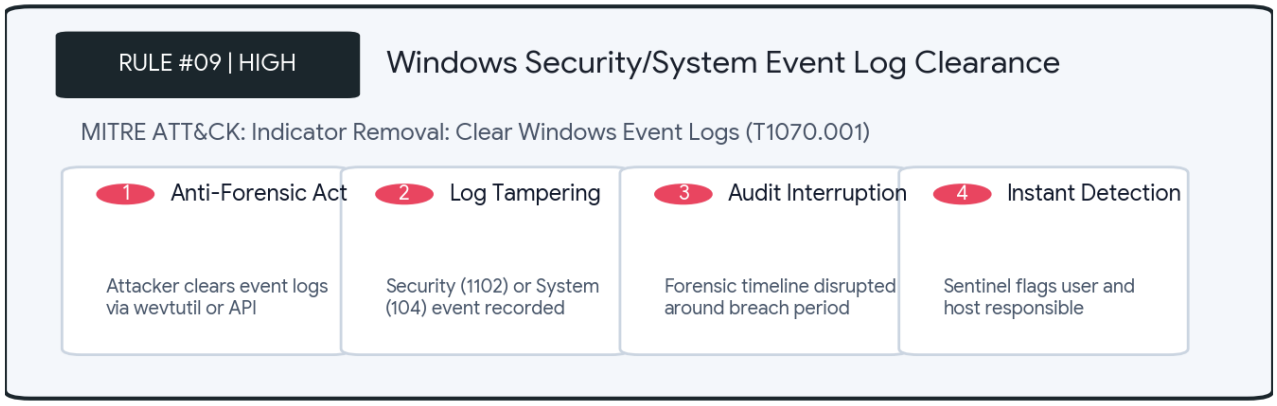
```
// Alert: User Added to Privileged Group
SecurityEvent
| where EventID in (4728, 4732)
| extend TargetGroup = tostring(TargetUserName),
        AddedMember = tostring(MemberName),
        SubjectUser = tostring(SubjectUserName)
| where TargetGroup in~ (
    "Administrators", "Domain Admins",
    "Enterprise Admins", "Remote Desktop Users"
  )
| project TimeGenerated, Computer, TargetGroup, AddedMember, SubjectUser
| extend MitreTactic = "PrivilegeEscalation", MitreTechnique = "T1098"
```

| | |
|---|---|
| **False Positives & Tuning** | Legitimate IT Helpdesk provisioning or automated JIT (Just-In-Time) access workflows. Correlate with approved change request IDs. |
| **SOC Triage & Response** | 1. Confirm authorization of SubjectUser and change ticket.<br>2. If unauthorized, revoke membership immediately.<br>3. Audit recent logon activity for AddedMember. |

# RULE #09 | Windows Security/System Event Log Clearance

| | | | |
|---|---|---|---|
| MITRE TACTIC | Defense Evasion | MITRE TECHNIQUE | T1070.001 |
| LOG SOURCE / TABLE | SecurityEvent | ALERT SEVERITY | High |

**Threat Scenario & Context:** Monitors intentional clearing of Windows Security (Event ID 1102) or System (Event ID 104) event logs, a hallmark post-exploitation tactic to eliminate forensic evidence.




### Production KQL Detection Logic

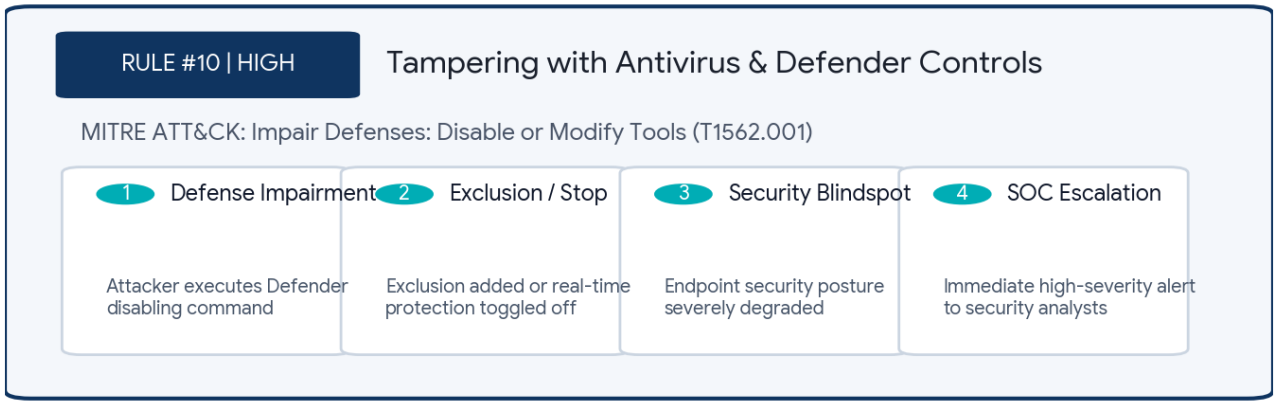
```
// Alert: Audit Log Cleared
SecurityEvent
| where EventID in (1102, 104)
| project TimeGenerated, Computer, Activity, SubjectUserName,
        SubjectDomainName
| extend MitreTactic = "DefenseEvasion", MitreTechnique = "T1070.001"
```

| | |
|---|---|
| **False Positives & Tuning** | Automated log rollover scripts or system maintenance during OS imaging/provisioning. |
| **SOC Triage & Response** | 1. Identify the user account in SubjectUserName.<br>2. Review preceding commands and logon sessions on Computer before the log wipe.<br>3. Isolate system. |

# RULE #10 | Tampering with Antivirus & Defender Controls

| | | | |
|---|---|---|---|
| MITRE TACTIC | Defense Evasion | MITRE TECHNIQUE | T1562.001 |
| LOG SOURCE / TABLE | DeviceProcessEvents | ALERT SEVERITY | High |

**Threat Scenario & Context:** Detects attempts to disable Windows Defender real-time monitoring, add path exclusions, unload minifilter drivers, or stop antivirus services via CLI/PowerShell.




### Production KQL Detection Logic

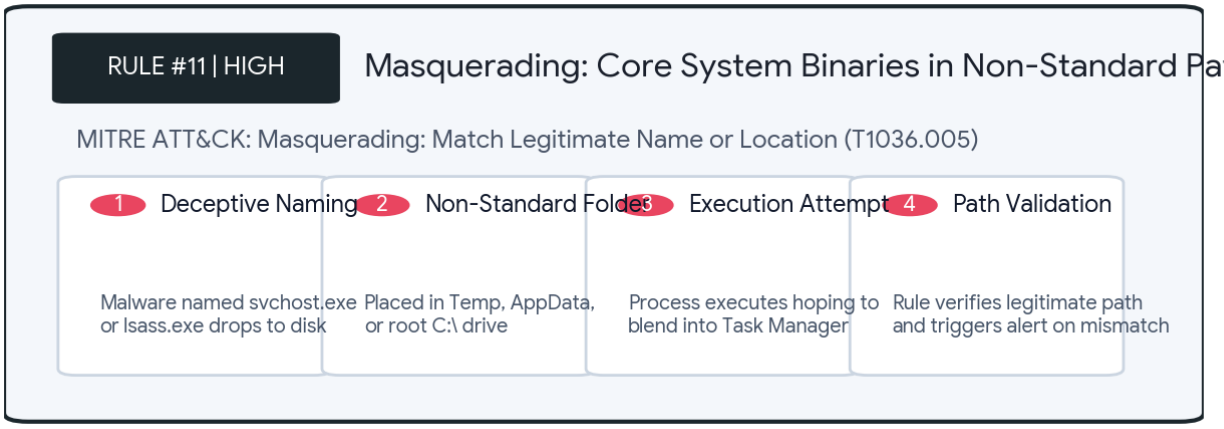
```
// Alert: Tampering with Antivirus / Defender Settings
DeviceProcessEvents
| where ProcessCommandLine has_any (
    "Set-MpPreference -DisableRealtimeMonitoring $true",
    "Add-MpPreference -ExclusionPath",
    "sc config WinDefend start= disabled",
    "net stop WinDefend"
  ) or (FileName =~ "fltmc.exe" and ProcessCommandLine has "unload")
| project TimeGenerated, DeviceName, AccountName, ProcessCommandLine,
          InitiatingProcessFileName
| extend MitreTactic = "DefenseEvasion", MitreTechnique = "T1562.001"
```

| | |
|---|---|
| **False Positives & Tuning** | DevOps setup scripts configuring folder exclusions for heavy compile builds. Verify change management tickets. |
| **SOC Triage & Response** | 1. Restore Defender settings (Set-MpPreference -DisableRealtimeMonitoring $false).<br>2. Identify the process that executed the command.<br>3. Full AV scan. |

# RULE #11 | Masquerading: Core System Binaries in Non-Standard Paths

| | | | |
|---|---|---|---|
| MITRE TACTIC | Defense Evasion | MITRE TECHNIQUE | T1036.005 |
| LOG SOURCE / TABLE | DeviceProcessEvents | ALERT SEVERITY | High |

**Threat Scenario & Context:** Detects critical system binaries (e.g., svchost.exe, lsass.exe, explorer.exe) executing outside their legitimate directory structures (System32, SysWOW64, WinSxS).




### Production KQL Detection Logic

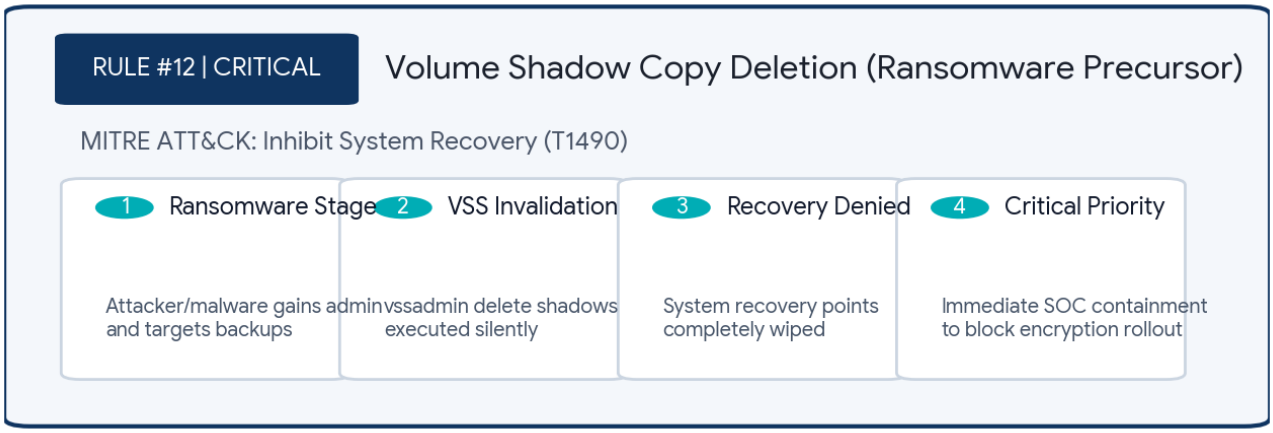
```
// Alert: Core System Binary Running Outside System32
DeviceProcessEvents
| where FileName in~ (
    "svchost.exe", "lsass.exe", "smss.exe", "services.exe",
    "csrss.exe", "explorer.exe"
  )
| where not(FolderPath has_any (
    @"C:\Windows\System32", @"C:\Windows\SysWOW64", @"C:\Windows\WinSxS"
  ))
| project TimeGenerated, DeviceName, FileName, FolderPath,
          ProcessCommandLine, AccountName
| extend MitreTactic = "DefenseEvasion", MitreTechnique = "T1036.005"
```

| | |
|---|---|
| **False Positives & Tuning** | Extremely low. Very few legitimate applications name their binaries after protected core Windows services. |
| **SOC Triage & Response** | 1. Terminate process immediately.<br>2. Collect binary hash and quarantine file.<br>3. Review persistence mechanisms and network connections. |

# RULE #12 | Volume Shadow Copy Deletion (Ransomware Precursor)

| | | | |
|---|---|---|---|
| MITRE TACTIC | Impact | MITRE TECHNIQUE | T1490 |
| LOG SOURCE / TABLE | DeviceProcessEvents | ALERT SEVERITY | Critical |

**Threat Scenario & Context:** Detects commands commonly utilized by ransomware operators (vssadmin, wbadmin, wmic, bcdedit) to delete volume shadow copies and disable boot recovery before encryption.




### Production KQL Detection Logic

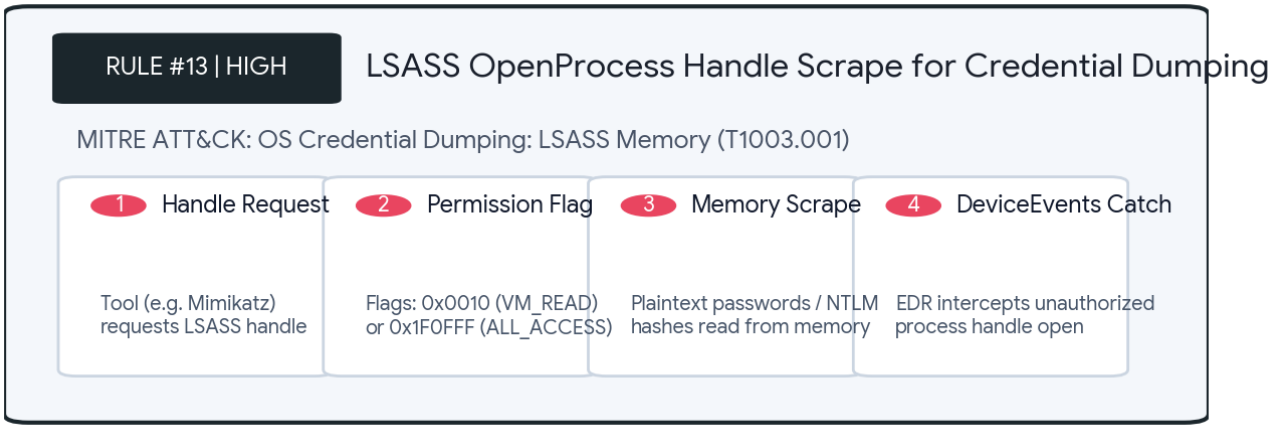
```
// Alert: VSS and Recovery Invalidation
DeviceProcessEvents
| where (FileName =~ "vssadmin.exe" and ProcessCommandLine has_all ("delete", "shadows"))
    or (FileName =~ "wbadmin.exe" and ProcessCommandLine has "delete")
    or (FileName =~ "wmic.exe" and ProcessCommandLine has_all ("shadowcopy", "delete"))
    or (FileName =~ "bcdedit.exe" and ProcessCommandLine has_any (
       "recoveryenabled No", "bootstatuspolicy ignoreallfailures"
    ))
| project TimeGenerated, DeviceName, FileName, ProcessCommandLine,
        InitiatingProcessFileName
| extend MitreTactic = "Impact", MitreTechnique = "T1490"
```

| | |
|---|---|
| **False Positives & Tuning** | Legitimate enterprise backup software purging outdated snapshots prior to creating full images. Verify backup service account. |
| **SOC Triage & Response** | 1. CRITICAL: Isolate device from network immediately.<br>2. Check for active bulk file encryption/renaming.<br>3. Engage Incident Response. |

# RULE #13 | LSASS OpenProcess Handle Scrape for Credential Dumping

| | | | |
|---|---|---|---|
| MITRE TACTIC | Credential Access | MITRE TECHNIQUE | T1003.001 |
| LOG SOURCE / TABLE | DeviceEvents | ALERT SEVERITY | High |

**Threat Scenario & Context:** Detects unauthorized processes requesting memory read permissions (PROCESS_VM_READ / PROCESS_ALL_ACCESS) against the Local Security Authority Subsystem Service (LSASS).




### Production KQL Detection Logic

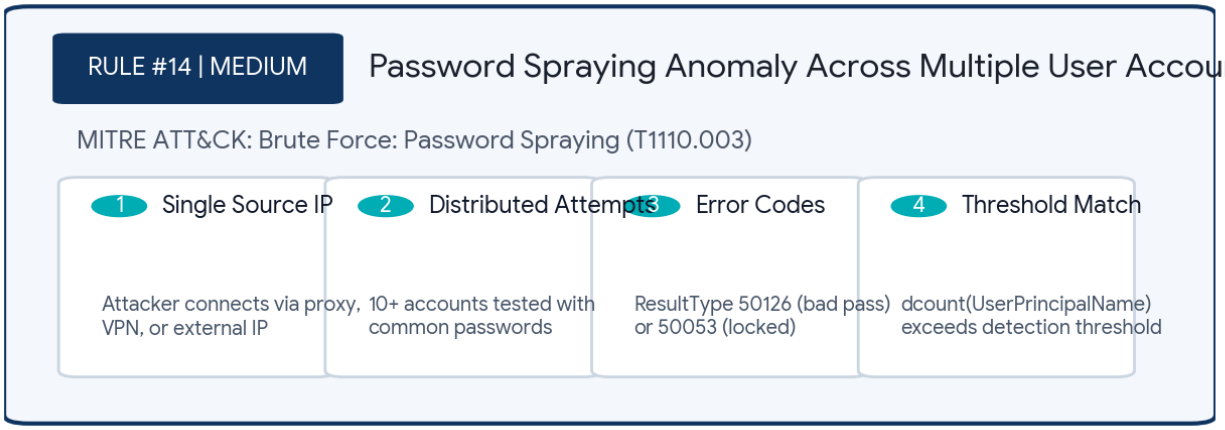
```
// Alert: Suspicious OpenProcess on LSASS
DeviceEvents
| where ActionType == "OpenProcessHandle"
| where FileName =~ "lsass.exe"
| where not(InitiatingProcessFileName in~ (
    "csrss.exe", "services.exe", "MsMpEng.exe", "svchost.exe"
  ))
| where AdditionalFields has_any ("0x0010", "0x1f0fff")
| project TimeGenerated, DeviceName, InitiatingProcessFileName,
          InitiatingProcessCommandLine
| extend MitreTactic = "CredentialAccess", MitreTechnique = "T1003.001"
```

| | |
|---|---|
| **False Positives & Tuning** | Third-party endpoint protection agents, IT monitoring tools, or password managers. Whitelist approved vendor binaries. |
| **SOC Triage & Response** | 1. Review InitiatingProcessFileName and hash.<br>2. Reset passwords for any accounts logged into that endpoint.<br>3. Check for lateral movement. |

# RULE #14 | Password Spraying Anomaly Across Multiple User Accounts

| | | | |
|---|---|---|---|
| MITRE TACTIC | Credential Access | MITRE TECHNIQUE | T1110.003 |
| LOG SOURCE / TABLE | SigninLogs | ALERT SEVERITY | Medium |

**Threat Scenario & Context:** Identifies an external or internal IP address attempting authentication against a high distinct count of enterprise user accounts within a 1-hour window using invalid credentials.




### Production KQL Detection Logic

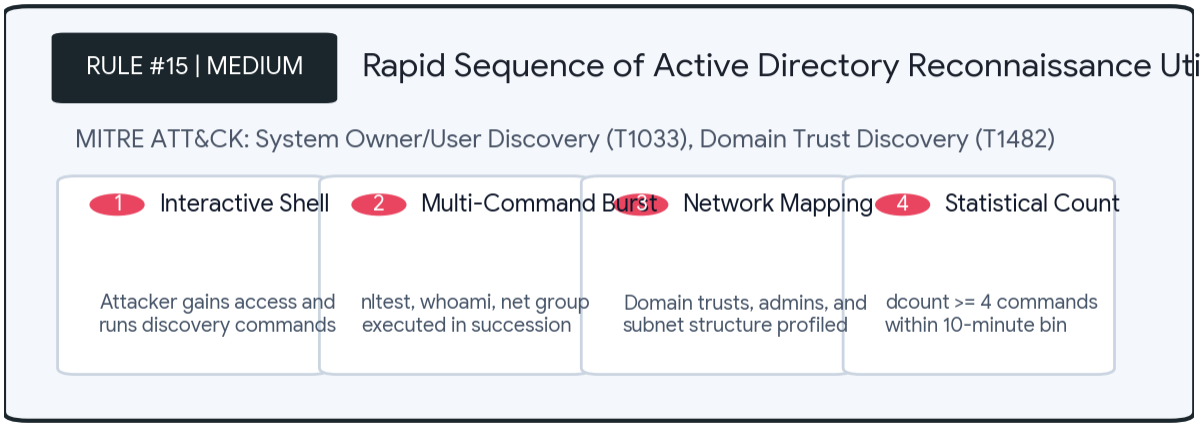
```
// Alert: Password Spray Pattern Detected
SigninLogs
| where TimeGenerated >= ago(1h)
| where ResultType in (50126, 50053) // Invalid password or Account locked
| summarize FailedUserCount = dcount(UserPrincipalName),
          TargetedUsers = make_set(UserPrincipalName, 10)
          by IPAddress
| where FailedUserCount >= 10
| project IPAddress, FailedUserCount, TargetedUsers
| extend MitreTactic = "CredentialAccess", MitreTechnique = "T1110.003"
```

| | |
|---|---|
| **False Positives & Tuning** | Misconfigured enterprise identity federation proxy or internal SSO gateway routing user failures under a single NAT gateway IP. |
| **SOC Triage & Response** | 1. Check IP reputation on IPQualityScore / AbuseIPDB.<br>2. Verify if any sign-in succeeded from the same IP (ResultType == 0).<br>3. Block IP on firewall. |

# RULE #15 | Rapid Sequence of Active Directory Reconnaissance Utilities

| | | | |
|---|---|---|---|
| MITRE TACTIC | Discovery | MITRE TECHNIQUE | T1482 |
| LOG SOURCE / TABLE | DeviceProcessEvents | ALERT SEVERITY | Medium |

**Threat Scenario & Context:** Detects rapid bursts of native discovery utilities (nltest, net, whoami, dsquery, klist) executed within a 10-minute window on a single endpoint, signalling active reconnaissance.




### Production KQL Detection Logic

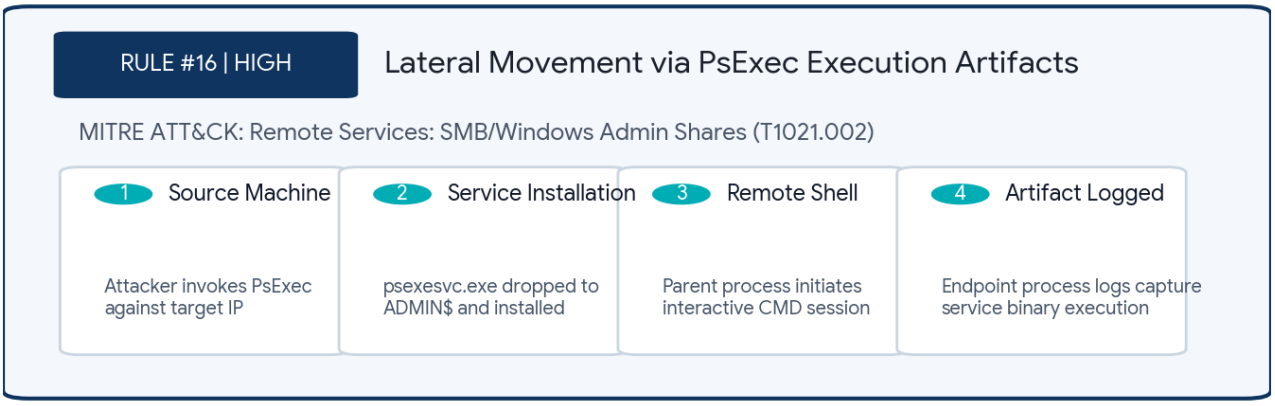
```
// Alert: Rapid Sequence of Reconnaissance Utilities
DeviceProcessEvents
| where TimeGenerated >= ago(30m)
| where FileName in~ (
    "nltest.exe", "net.exe", "whoami.exe", "dsquery.exe",
    "klist.exe", "systeminfo.exe", "ipconfig.exe"
  )
| summarize CmdCount = dcount(FileName),
          CommandsRun = make_set(ProcessCommandLine)
          by DeviceName, AccountName, bin(TimeGenerated, 10m)
| where CmdCount >= 4
| extend MitreTactic = "Discovery", MitreTechnique = "T1033"
```

| | |
|---|---|
| **False Positives & Tuning** | System administrators troubleshooting network/domain connectivity manually or running custom diagnostics scripts. |
| **SOC Triage & Response** | 1. Contact AccountName to confirm if troubleshooting was active.<br>2. Check preceding process lineage (parent process).<br>3. Monitor for lateral movement. |

# RULE #16 | Lateral Movement via PsExec Execution Artifacts

| | | | |
|---|---|---|---|
| MITRE TACTIC | Lateral Movement, Execution | MITRE TECHNIQUE | T1021.002 |
| LOG SOURCE / TABLE | DeviceProcessEvents | ALERT SEVERITY | High |

**Threat Scenario & Context:** Detects the presence and execution of PsExec service binaries (psexesvc.exe) and command lines, indicating remote code execution across administrative SMB shares (ADMIN$).




### Production KQL Detection Logic

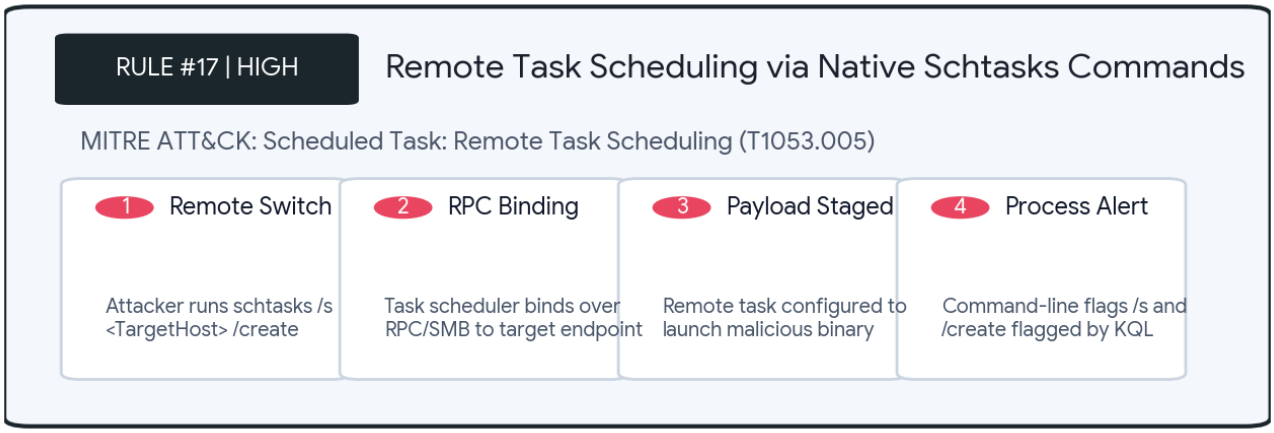
```
// Alert: PsExec Execution Artifact Detected
DeviceProcessEvents
| where FileName =~ "psexesvc.exe"
    or InitiatingProcessFileName =~ "psexesvc.exe"
    or ProcessCommandLine has "psexec"
| project TimeGenerated, DeviceName, AccountName, FileName,
          ProcessCommandLine, InitiatingProcessFileName
| extend MitreTactic = "LateralMovement", MitreTechnique = "T1021.002"
```

| | |
|---|---|
| **False Positives & Tuning** | Authorized IT sysadmin maintenance or vulnerability scanners (Nessus, Qualys). Baseline known admin source IP ranges. |
| **SOC Triage & Response** | 1. Trace originating IP from SMB session logs (Event ID 4624/5140). 2. Inspect command executed via PsExec. 3. Isolate source and destination. |

# RULE #17 | Remote Task Scheduling via Native Schtasks Commands

| | | | |
|---|---|---|---|
| MITRE TACTIC | Lateral Movement, Persistence | MITRE TECHNIQUE | T1053.005 |
| LOG SOURCE / TABLE | DeviceProcessEvents | ALERT SEVERITY | High |

**Threat Scenario & Context:** Detects the use of schtasks.exe with the /s parameter to remotely register scheduled tasks on remote network computers using native RPC/SMB protocols.




### Production KQL Detection Logic

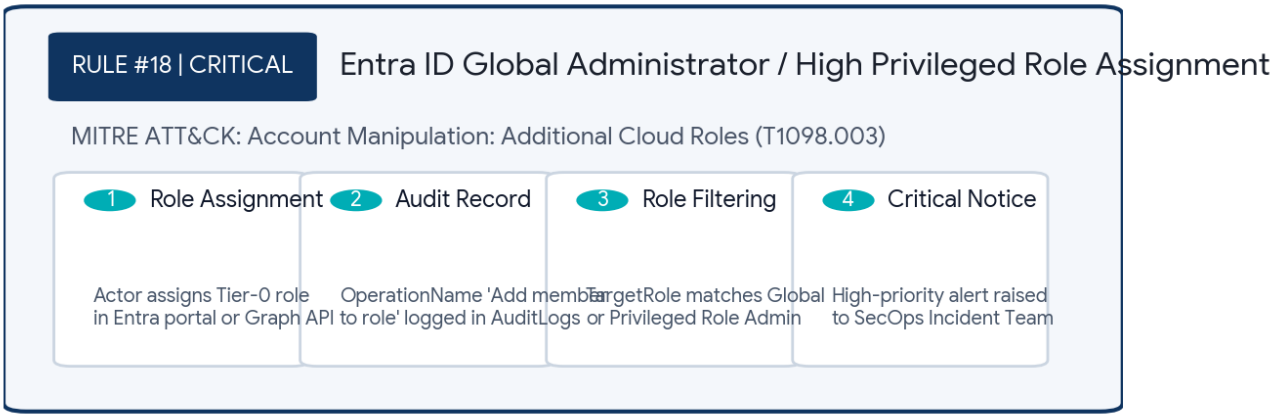
```
// Alert: Remote Task Scheduling
DeviceProcessEvents
| where FileName =~ "schtasks.exe"
| where ProcessCommandLine has "/s " and ProcessCommandLine has "/create"
| project TimeGenerated, DeviceName, AccountName, ProcessCommandLine,
        InitiatingProcessCommandLine
| extend MitreTactic = "LateralMovement", MitreTechnique = "T1053.005"
```

| | |
|---|---|
| **False Positives & Tuning** | Legacy central software deployment scripts. Modern environments utilize Intune, SCCM, or Ansible. |
| **SOC Triage & Response** | 1. Identify the target host passed in the /s argument. 2. Verify account privileges. 3. Check target host task scheduler logs (EID 106). |

# RULE #18 | Entra ID Global Administrator / High Privileged Role Assignment

| | | | |
|---|---|---|---|
| MITRE TACTIC | Privilege Escalation, Persistence | MITRE TECHNIQUE | T1098.003 |
| LOG SOURCE / TABLE | AuditLogs | ALERT SEVERITY | Critical |

**Threat Scenario & Context:** Audits Microsoft Entra ID (Azure AD) administrative role grants, alerting immediately whenever an identity is granted Tier-0 directory roles such as Global Administrator.




**Production KQL Detection Logic**

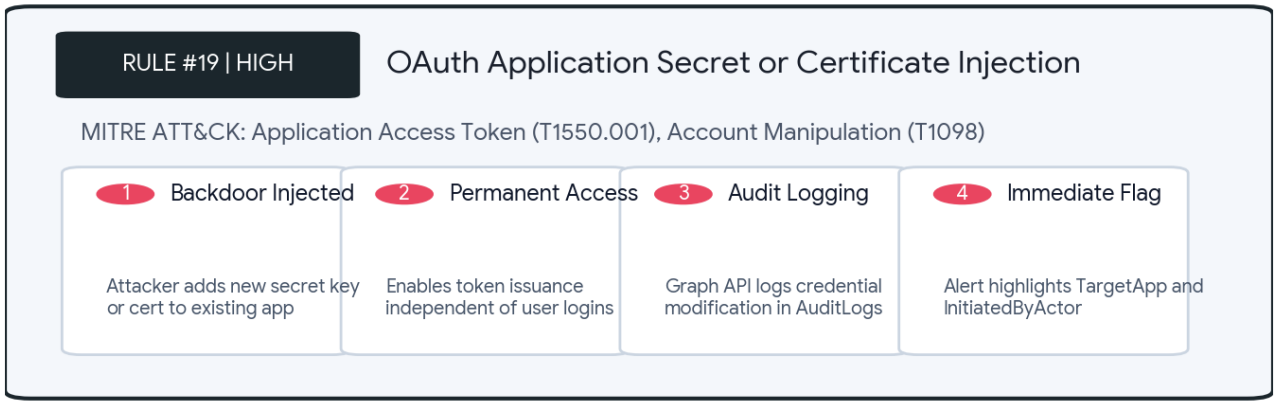
```
// Alert: Global Administrator or High Privileged Role Assigned
AuditLogs
| where OperationName in~ (
    "Add member to role", "Add eligible member to role"
  )
| extend TargetRole = tostring(parse_json(
    tostring(TargetResources[0].modifiedProperties)
  )[0].newValue)
| where TargetRole has_any (
    "Global Administrator", "Privileged Role Administrator",
    "Security Administrator"
  )
| extend TargetUser = tostring(TargetResources[0].userPrincipalName)
| extend InitiatedByActor = tostring(InitiatedBy.user.userPrincipalName)
| project TimeGenerated, TargetRole, TargetUser, InitiatedByActor, Result
| extend MitreTactic = "PrivilegeEscalation", MitreTechnique = "T1098.003"
```

| | |
|---|---|
| **False Positives & Tuning** | Approved PIM (Privileged Identity Management) elevations. Filter out legitimate PIM workflows by verifying InitiatedBy app IDs. |
| **SOC Triage & Response** | 1. Validate with Cloud Architecture team if change was scheduled. 2. Check initiating actor MFA and IP reputation. 3. Revoke role assignment if unapproved. |

# RULE #19 | OAuth Application Secret or Certificate Injection

| | | | |
|---|---|---|---|
| MITRE TACTIC | Persistence, Defense Evasion | MITRE TECHNIQUE | T1098 |
| LOG SOURCE / TABLE | AuditLogs | ALERT SEVERITY | High |

**Threat Scenario & Context:** Detects the addition of cryptographic credentials (client secrets or certificates) to existing Entra ID Enterprise Applications, a primary persistence mechanism for cloud adversaries.




### Production KQL Detection Logic

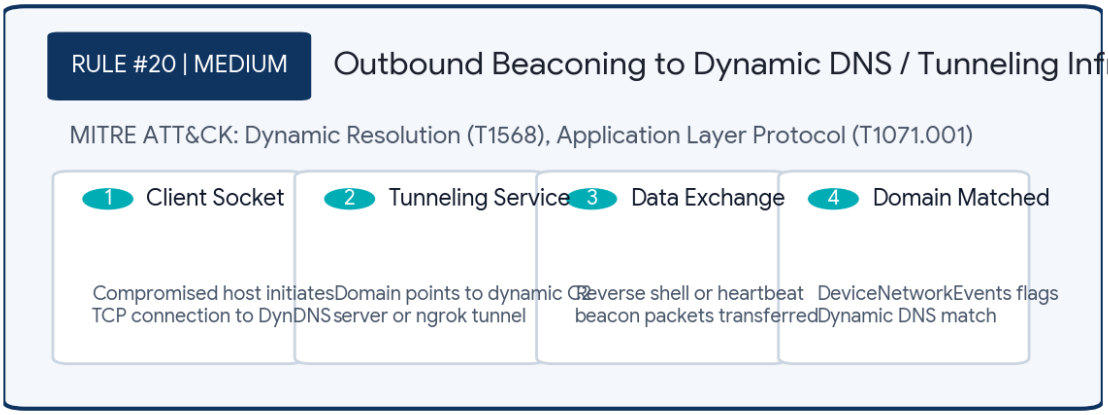
```
// Alert: Secret/Certificate Added to Application
AuditLogs
| where OperationName has_any (
    "Update application – Certificates and secrets management",
    "Add service principal credentials"
  )
| extend TargetApp = tostring(TargetResources[0].displayName)
| extend InitiatedByActor = tostring(InitiatedBy.user.userPrincipalName)
| project TimeGenerated, OperationName, TargetApp, InitiatedByActor, Result
| extend MitreTactic = "Persistence", MitreTechnique = "T1098"
```

| | |
|---|---|
| **False Positives & Tuning** | DevOps automated credential rotation pipelines (e.g. Terraform, KeyVault automation). Baseline automation Service Principals. |
| **SOC Triage & Response** | 1. Check TargetApp permissions (e.g., Mail.ReadWrite, Directory.ReadWrite.All). 2. Verify who generated the secret. 3. Invalidate compromised secret. |

# RULE #20 | Outbound Beaconing to Dynamic DNS / Tunneling Infrastructure

| | | | |
|---|---|---|---|
| MITRE TACTIC | Command and Control | MITRE TECHNIQUE | T1071.001 |
| LOG SOURCE / TABLE | DeviceNetworkEvents | ALERT SEVERITY | Medium |

**Threat Scenario & Context:** Monitors internal endpoints initiating successful network connections to Dynamic DNS providers or reverse tunneling services commonly leveraged for C2 traffic and data exfiltration.




### Production KQL Detection Logic

```
// Alert: Network Connection to Dynamic DNS Domain
let DynDNSDomains = dynamic([
    "duckdns.org", "no-ip.com", "ddns.net", "hopto.org",
    "zapto.org", "sytes.net", "ngrok.io", "portmap.io"
]);
DeviceNetworkEvents
| where ActionType == "ConnectionSuccess"
| where RemoteUrl has_any (DynDNSDomains)
| project TimeGenerated, DeviceName, InitiatingProcessFileName,
          RemoteUrl, RemoteIP, RemotePort
| extend MitreTactic = "CommandAndControl", MitreTechnique = "T1568"
```

| | |
|---|---|
| **False Positives & Tuning** | Developers using ngrok.io for authorized local webhook testing. Require security exception requests for developer tunneling tools. |
| **SOC Triage & Response** | 1. Identify the InitiatingProcessFileName and PID.<br>2. Terminate the connection and isolate the host.<br>3. Inspect internal ports exposed via tunnel. |